

Akhil Basavanapalli

Data Engineer

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Summary

- 4+ years of IT experience in software analysis, design, development, testing, and implementation with **Python/PySpark**, **SQL**, and **Azure** technologies.
- Proficient in Azure services including **Data Factory**, **Synapse Analytics**, **Data Bricks**, **Logic Apps**, and **Automation Accounts**.
- Implemented Lambda architectures using **Azure Data Lake**, **Azure SQL Server**, **Azure ML**, and **Power BI**.
- Skilled in using Developer Toolkits like **Anaconda**, **Jupyter Notebooks**, and **PyCharm**.
- Experienced across all SDLC phases: **Analysis**, **Design**, **Implementation**, **Testing**, and **Maintenance**.
- Utilized **Azure Fabric** for migrating notebooks and pipelines, maintaining data warehouses in **OneLake**.
- Worked with **Azure Data Lake** and various file formats (csv, parquet, orc, delta).
- Documented business, process, and data flows using **Lucid Chart**.
- Extensive experience in developing **stored procedures**, **functions**, **Views** and **Triggers**, **Complex queries** using **SQL Server**, **TSQL** and **Oracle PL/SQL**.
- Designed data models, provided customer support, troubleshooting, and database modifications.
- Enhanced Terraform scripts for AWS services including **CloudFront**, **RDS**, **EC2**, and **S3**.
- Expertise in **Snowflake** for creating and maintaining tables and views.
- Hands-on with ML/AI algorithms using **R** and **Python** libraries for predictive analytics.
- Knowledgeable in Azure cloud IaaS/PaaS services, **Spark SQL**, **NoSQL** databases (**Cassandra**, **MongoDB**), and data warehouses (**Azure Synapse**, **Snowflake**).
- Experienced with AWS Cloud services (**EC2**, **SNS**, **SQS**, **S3**, **EBS**, **RDS**, **RedShift**, **Lambda**, **CloudWatch**), and big data ecosystems (**Hadoop**, **Spark**, **Hive**).
- Developed **Power BI** reports, effective dashboards, and conducted comprehensive code reviews for software projects.

Skill

Cloud Platform	Microsoft Fabric, Azure Data Factory, Azure Synapse Analytics, Azure Data Bricks, Azure Data Lake, Azure Blob Storage, Amazon AWS, EC2, MS Azure, Azure SQL Database, Azure SQL Data Warehouse, Azure Analysis Services, AWS Lambda, SNS, SQS, Amazon Redshift, RDS, EC2, S3 Bucket.
Programming Language	Python 3.6, Pyspark, Scala, SQL, PL/SQL, DB2
Databases	SQL, Oracle
Operating Systems	Windows 10/8/7, Linux
IDE and Tools	Anaconda, Jupyter Notebook
SDLC	Agile, Waterfall
Methodologies	
Version Control	GIT, Azure Devops
Reporting tool	PowerBI, Tableau
Other Tools	Visual Studio 2010, Business Intelligence Studio 2008, Power BI, SQL Server Integration Services (SSIS) 2005/2008, SQL, SQL Server Reporting Services (SSRS) 2008, SQL Server 2008 R2, Postman, PowerShell.

Experience

Data Engineer | Synchrony Financial, CT USA | Feb 2023 to Current

- Develop **Synapse** and **ADF Data copy pipelines**, Data flows, and **Pyspark** notebooks using **Data Bricks** and **Synapse Analytics**.
- Create and manage data warehouses and Data copy pipelines using **Pyspark** notebooks in **Microsoft Fabric**.
- Automate **ETL/ELT processes** using **Pyspark** scripts from multiple sources.
- Develop stored procedures for data warehouse management and scheduling.
- Create notebooks to read data from **Salesforce** and **SugarCRM** using **SOQL/SOSL** for **data warehouse** management.
- Develop ETL pipelines to copy data from **Business Central** and **EDH** using **REST API methods**.
- Use **Azure Logic Apps** to copy **Tempo Data** from **Jira** to **Business Central** on a bi-weekly schedule.
- Create and maintain **Power BI** reports in **Microsoft Fabric** for **Accounting** and **Financials**.
- Create pipelines and transformation techniques in **Notebooks** in **Microsoft Fabric**.
- Develop and maintain data solutions in **OneLake** within **Microsoft Fabric**.
- Utilize **Pyspark** scripts with **SQL** and **RDD** in **Spark** for data analysis and storage in **Azure Blob Storage** and **S3**.

- Perform data migration from AWS S3 to Snowflake using Copy and conduct data validation.
- Use **Snowpark API** and Pandas for data wrangling and streamline data via Python APIs.
- Perform data transformation using Snowflake and **AWS Glue**.
- Develop ETL pipelines using complex custom **SQL** in **Python**.
- Architect new solutions for reading data from Jira cloud and generating Power BI reports.
- Implement business logic using **Spark SQL** and **Python programming**.
- Enhance Hive performance through sequence files, map-side joins, bucketing, and partitioning.
- Develop and maintain **AWS Lambda functions**, **DynamoDB** tables, **secret manager**, and SQL queues.
- Collaborate with Finance teams and contractors and deploy new developments into production.

Data Engineer | Informative Web Solutions, India | May 2018 to Jul 2021

- Data Engineering: Extracted, transformed, and loaded data from source systems to Azure Data Storage using Azure Data Factory, T-SQL, Spark SQL, and U-SQL Azure Data Lake Analytics.
- Data Warehousing: Architected the data flow for consolidating legacy data warehouses into Azure Data Lake Storage (ADLS) and designed the ELT strategy into an Analytics Sandbox and EDW.
- Cloud Migration: Migrated on-premises data warehouses to **Azure Synapse** and developed **ETL processes** using T-SQL and **SSIS**.
- Data Pipelines: Built data pipelines on Azure Data Factory using SAS or **PySpark**, and created reusable ETL processes with Azure Synapse Pipelines.
- Programming and Scripting: Developed, tested, and maintained **SAS** and Python programs, automated workloads with PowerShell and Azure CLI scripts.
- Data Processing: Loaded and transformed large datasets (XML, JSON, Avro, Parquet) and scheduled Elastic jobs for TSQL procedures.
- Reporting and Visualization: Designed reports and visualizations using SAS Visual Analytics and Power BI, integrated custom visuals in Power BI desktop.
- Automation: Automated recurring SQL database activities with **PowerShell**, developed templates for Azure infrastructure using Terraform.
- Data Analysis: Performed data mining and predictive analytics using AI and ML techniques, developed complex SQL queries for Power BI and SSRS reports.
- Infrastructure Management: Built and deployed data pipelines on Azure for AI & ML capabilities, integrated **Azure Log Analytics** with VMs.
- ETL Development: Created SSIS packages for data transfer, used SSIS variables for dynamic execution, and wrote complex calculated measures with DAX.
- Web Crawling: Utilized Python modules (urllib, requests) for web crawling, and applied ML techniques like **clustering**, **regression**, and **classification**.
- Power BI Integration: Embedded Power BI reports in SharePoint, managed report access, and designed cloud-based solutions with Azure SQL and Elastic pools.
- Big Data Technologies: Developed Spark applications with PySpark on Databricks, optimized Spark applications for performance, and created Hive tables and queries.
- Production Support: Provided 24x7 production support, created pipelines in ADF for data extraction and transformation, and managed Azure Data Lake storage.

Defense Research and Development Organization (DRDO) | Internships | Sep 2017 to Apr 2018

- Software Development: Assist in developing, debugging, and troubleshooting software applications for defense purposes, collaborating with senior developers and engineers.
- Testing and Quality Assurance: Participate in unit testing, integration testing, and system validation, documenting test cases and results to support quality assurance.
- Research and Analysis: Conduct research on emerging technologies, analyze data and software architectures, and identify potential improvements for existing systems.
- Cybersecurity: Learn and apply cybersecurity principles, identify and mitigate security vulnerabilities, conduct code reviews, and implement security controls.
- Documentation and Compliance: Document software configurations and test results to support compliance requirements and ensure the security and integrity of defense-related systems.

Education

Master of Science in Computer Science | Kennesaw State University, Atlanta GA, Aug 2021 - Dec 2022.

Bachelor of Engineering in Information Technology | Sathyabama University, India, 2014 – 2018.

Certifications

Microsoft Certified Expert | Credential ID: 80C25299E9611C7E